

CONSOLIDATION — BST program state after the GR/ Λ /CFS/CP frontier arc (2026-08-08 $\rightarrow$ 09)

Keeper, 2026-08-09. Consolidating what the two-day frontier arc established, at true tier, + the papers to write + the broader backlog. The real product of the arc: every claim landed at its tier and the discipline caught ~ 9 over-reads (several mine) before any banked.

What the frontier arc BANKED (at tier)

- QM from D_{IV}^5 — Derived, 10/10 axioms, ship-ready (#67).
- Gravity is emergent, no graviton, holographic — Framework (Jacobson/Sakharov/AdS-CFT anchored); never “derived Einstein.”
- Dark energy — SIGN forced sub-Poisson (Bernstein/complete-monotonicity), rate = C_2 (survived 3 rescue attempts), homeostat/relaxing-to-conformal mechanism; survives the DESI DR2 radial BAO (real data, no crossing); shares Λ CDM’s $\sim 3\sigma$ tension on BAO+CMB; differentiator = a monotone one-sign power-law residual (K1298). Λ stays Structural.
- #79 (is D_{IV}^5 forced) $\rightarrow$ “forced modulo ONE datum, the generation count” — GR-level, two legs: (A) rank=2 by generation count via source-verified Korányi-Wolf ($r+1$ strata, Wolf1972/Loos1977, E7 excluded); (B) D_{IV}^5 within rank-2 via census $N_c = \text{rank}^2 - 1$ (T1829) + color-dim (K989/K1055). The Tegmark headline (#31/#25).
- CP violation FORCED (K1303) — both legs forced: misalignment (up-saturation $\neq$ down-feed-down, different mechanisms) + the complex twist ($n_C=5$ odd $\rightarrow$ $\text{Spin}(5)=\text{Sp}(2)$ quaternionic/pseudoreal spinor, signature 5 mod 8, no real structure $\rightarrow$ non-removable $\sqrt{-1}$; source-verified); $\det[H_u, H_d] \neq 0$, J rephasing-invariant. Existence Structural; magnitude OFF. Banks on 2 blind confirmations.
- The CFS/Finster bridge — the closed chain = a 4×4 eigenvalue test on the (2,2) spin fiber; the same forced complex twist forces the light-cone (F870/F871), so CP + space-from-time + the credential are ONE geometric fact. Thesis: BST FORCES what CFS ALLOWS (the physical grounding Finster’s general framework leaves free). Gated behind the outreach pass.
- Clean-repo — Millennium sector made honest (banners + 17 PDFs rebuilt); Obata retraction closed. Flagship wording (bst_this_is.md, README Wyler) = Casey’s call.

The through-line (the consolidated theory)

ONE geometry D_{IV}^5 , forced modulo one empirical datum (the generation count). Everything is a reading of one operator (linear algebra on D_{IV}^5). The oddness of the integers is the engine: odd $N_c=3 \rightarrow$ color-count $\rightarrow \sin^2\theta_W$, the $\mathbb{Z}_3$; odd $n_C=5 \rightarrow$ quaternionic spinor $\rightarrow$ CP; the whole SM + gravity + QM as facets of the one commitment geometry. BST = Tegmark’s MUH + a selection principle.

Papers to write (consolidation vehicles)

1. Forcing + Evidence / Tegmark paper (#31/#25) — the flagship: forced-modulo-one-datum + CP-forced + DE + the CFS bridge. Keeper + Lyra.
2. The one-page minimal generating set (#66) — one object + ~ 5 operations, every observable a reading. Grace.
3. QM-from- D_{IV}^5 (#67) — ship-ready; finalize.
4. The CFS/Finster common-ground note — held/gated (the linear-algebra recast + “BST forces what CFS allows”).

The broader backlog (keep momentum — the bulk of the program)

- Mixing sector / the 7 parameters (#68 FK overlap, #69 O7, #77 ordered-product) — now that CP is forced, the CKM/PMNS finish line is ripe (the phase is a forced input).
- Charged leptons (#36/#37) — the tau seat, the lepton module.

- Strong sector (#17/#21) — β -function/running $\rightarrow$ YM mass gap.
- F85 geodesic condensate (#78) — the up-ordering (just proved load-bearing for CP).

— Keeper, consolidation 2026-08-09. The arc is banked and honest; the papers are the consolidation vehicles; the broader SM-parameter backlog keeps the momentum. Nothing pushed.